

LeafScan- Plant Disease Detection

A Synopsis for Minor Project

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

BY

Radhika Gurjar EN23CS301813,

Priyanshi Bijghawane EN23CS301789,

Priyanshi Mahajan EN23CS301790

Under the Guidance of
Prof. Swati Tahiliani , Prof. Laxmi Kag



MEDICAPS
U N I V E R S I T Y

Department of Computer Science & Engineering
Faculty of Engineering
MEDICAPS UNIVERSITY, INDORE- 453331

Jan – June 2026

Introduction

Artificial Intelligence and image processing techniques are increasingly transforming modern agricultural practices. Crop health monitoring, in particular, relies heavily on visual inspection of plant leaves to identify diseases. Plant diseases are one of the leading causes of crop yield loss worldwide and pose a serious threat to global food security. According to reports published by the Food and Agriculture Organization (FAO), a significant portion of agricultural production is lost each year due to plant diseases and delayed diagnosis. Early detection of plant diseases can greatly reduce crop damage and improve agricultural productivity.

However, farmers—especially in rural and remote areas—often lack access to agricultural experts and plant pathologists. Traditional disease identification methods depend on manual observation, which is time-consuming, costly, and subjective.

The system utilizes publicly available and field-based plant disease datasets such as PlantVillage and real-world annotated field datasets for training and evaluation. The objective of LeafScan is to assist farmers and agricultural professionals by providing an automated, accessible, and reliable tool for early plant disease detection and crop protection.

Literature Review

Image processing and Artificial Intelligence techniques have been widely applied in agricultural disease detection systems to support farmers and improve crop management. Research studies show that computer-aided plant disease diagnosis systems can significantly enhance detection accuracy while reducing human dependency and error. Publicly available datasets such as **PlantVillage** have played an important role in advancing research by providing labeled leaf images for training and testing machine learning and deep learning models. The PlantVillage dataset includes images of healthy and diseased leaves captured under controlled laboratory conditions and has been extensively used in academic studies.

Several traditional image processing techniques have been explored for plant disease detection before the rise of deep learning models. These techniques focus on segmenting the leaf area, extracting visual features, and classifying diseases based on those features. Commonly used methods include:

- Color-based thresholding to separate diseased regions
- Edge detection techniques for identifying lesion boundaries
- Morphological operations for noise removal and shape enhancement
- Texture feature extraction using Gray Level Co-occurrence Matrix (GLCM)
- Statistical analysis of color and texture patterns

More recently, Convolutional Neural Networks (CNNs) have gained popularity due to their ability to automatically learn discriminative features from images without manual feature extraction

- Problem Definition

Plant disease detection depends on analyzing visual characteristics of leaf images such as:

- Color variations on the leaf surface
- Presence of spots, blight, or lesions
- Shape and size of infected regions
- Texture patterns and surface irregularities

Manual identification of plant diseases is subjective and highly dependent on the farmer's experience or the availability of agricultural experts. In rural and resource-limited regions, access to plant pathologists and agricultural advisory services is limited. Additionally, similar visual symptoms across different diseases often lead to misdiagnosis, delayed treatment, and increased crop loss.

Therefore, there is a need for an automated system that can:

- Detect diseased regions from plant leaf images
- Extract meaningful visual features related to disease symptoms

- Classify plant diseases accurately using image processing and machine learning techniques

The problem addressed in this project is:

To develop an image-based plant disease detection and classification system, **LeafScan**, that assists farmers in early identification of crop diseases by analyzing leaf images using structured image processing and learning algorithms, thereby enabling timely intervention and reducing agricultural losses

Objectives

The objectives of **LeafScan** are:

1. To study various types of plant leaf diseases and their visual symptoms.
2. To collect and analyze plant leaf image datasets for disease detection.
3. To implement image preprocessing techniques to enhance leaf image quality.
4. To perform leaf region segmentation using image processing methods.
5. To extract color, texture, and shape features from leaf images.
6. To classify plant diseases using machine learning and deep learning algorithms.
7. To evaluate system performance using standard evaluation metrics such as accuracy and precision.
8. To develop a web-based prototype for academic and practical demonstration.

Methodology

The proposed methodology for **LeafScan** consists of the following stages:

Data Collection

Plant leaf images are collected from publicly available plant disease datasets such as:

- **PlantVillage Dataset**
- **Field-based plant disease image datasets**

These datasets contain labeled images of healthy and diseased leaves across multiple crop species and disease categories. The collected images represent variations in color, texture, and disease severity, which are essential for robust model training.

Image Preprocessing

Preprocessing is performed to enhance image quality and prepare the leaf images for further analysis. The following techniques are applied:

- Image resizing to maintain uniform input dimensions
- RGB to grayscale conversion (where required)
- Noise removal using median or Gaussian filtering
- Contrast enhancement using histogram equalization
- Background normalization to reduce lighting variations

These preprocessing steps follow standard digital image processing principles and help improve the accuracy of segmentation and classification.

Leaf Segmentation

Leaf segmentation is carried out to isolate the leaf region from the background. Segmentation techniques such as thresholding and edge-based methods are used to identify leaf boundaries. This step helps in focusing the analysis on the infected regions of the leaf.

Feature Extraction

After segmentation, meaningful visual features are extracted from the leaf images. These include:

- Color features to detect discoloration and spots
- Texture features using statistical methods such as GLCM
- Shape features to analyze the structure of infected areas

Disease Classification

The extracted features are used to classify plant diseases using machine learning and deep learning algorithms such as Convolutional Neural Networks (CNNs). Supervised learning is applied to distinguish between healthy and diseased leaves.

Performance Evaluation

The system performance is evaluated using standard metrics such as accuracy, precision, recall, and F1-score. The evaluation helps in assessing the effectiveness and reliability of the LeafScan system for plant disease detection.

Applications

LeafScan can be applied in:

- Early detection of plant diseases through leaf image analysis
- Smart agriculture and precision farming systems
- Decision-support systems for farmers and agricultural experts
- Mobile and web-based plant disease advisory platforms
- Academic and research-based agricultural studies

Limitations

- The system does not replace expert agricultural consultation or plant pathologists.
- Detection accuracy depends on the quality of the captured leaf images.
- Variations in lighting, background clutter, and overlapping leaves may affect results.
- Similar visual symptoms across different diseases may lead to misclassification.
- The system requires large, labeled datasets for effective training and performance.

Conclusion

This project presented **LeafScan**, an image-based plant disease detection system designed to assist in the early identification of crop diseases using leaf images. By integrating image processing techniques with machine learning and deep learning models, the system aims to provide an automated and reliable solution for plant disease classification. The use of preprocessing, segmentation, feature extraction, and classification stages enables effective analysis of leaf images and improves detection accuracy.

Journal References

- [1]. E. Moupojou *et al.*, "FieldPlant: A Dataset of Field Plant Images for Plant Disease Detection and Classification With Deep Learning," in *IEEE Access*, vol. 11, pp. 35398-35410, 2023, doi: 10.1109/ACCESS.2023.3263042.
- [2]. N. Nigar, H. Muhammad Faisal, M. Umer, O. Oki and J. Manappattukunnel Lukose, "Improving Plant Disease Classification With Deep-Learning-Based Prediction Model Using Explainable Artificial Intelligence," in *IEEE Access*, vol. 12, pp. 100005-100014, 2024, doi: 10.1109/ACCESS.2024.3428553.
- [3]. Ferentinos, Konstantinos P. "Deep learning models for plant disease detection and diagnosis." *Computers and electronics in agriculture* 145 (2018): 311-318.

Conference References

- [1]. H. Parmar and M. Rai, "Plant Leaf Disease Detection Using Multiple CNN Models," *2025 6th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)*, Goathgaun, Nepal, 2025, pp. 1136-1141, doi: 10.1109/ICMCSI64620.2025.10883506.

URL References

- [1.] <https://ieeexplore.ieee.org/document/11134206>
